

Discussion 20 — Complete Proofs

Linear Algebra (Lay) Reference Sheet

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Contents

Problem 1a: $\langle Ax, y \rangle = \langle x, Ay \rangle$ for Symmetric A

Proof Strategy

- Rewrite inner products as matrix products: $\langle u, v \rangle = u^T v$.
- Apply the transpose identity $(AB)^T = B^T A^T$ to $(Ax)^T$.
- Substitute $A^T = A$ (definition of symmetric).
- **Pattern:** Any time you need to “move” a matrix from one argument of an inner product to the other, transpose it and use symmetry.

Relevant Theorems & Definitions (Lay)

- **Def (Lay §6.4):** A is symmetric $\Leftrightarrow A^T = A$.
- **Thm (Lay §2.1, Thm 3):** $(AB)^T = B^T A^T$.
- **Inner product as matrix product:** $\langle u, v \rangle = u^T v$ for column vectors $u, v \in \mathbb{R}^n$.

Proof

Let A be $n \times n$ symmetric, and $x, y \in \mathbb{R}^n$.

$$\langle Ax, y \rangle = (Ax)^T y = x^T A^T y = x^T A y = \langle x, Ay \rangle.$$

Step 1: definition of inner product. Step 2: $(AB)^T = B^T A^T$. Step 3: $A^T = A$. Step 4: definition of inner product. ■

Problem 1b: $B^T AB$, $B^T B$, BB^T Are Symmetric

Proof Strategy

- To show M is symmetric, show $M^T = M$.
- Apply $(ABC)^T = C^T B^T A^T$ and substitute $A^T = A$.
- **Pattern:** Symmetry proofs always reduce to a single transpose computation. Never use dimension arguments alone — they confirm shape but not symmetry.

Relevant Theorems & Definitions (Lay)

- **Def (Lay §6.4):** M symmetric $\Leftrightarrow M^T = M$.
- **Thm (Lay §2.1, Thm 3):** $(ABC)^T = C^T B^T A^T$.
- **Note:** $(B^T)^T = B$.

Proof

Case 1: $B^T B$.

$$(B^T B)^T = B^T (B^T)^T = B^T B. \quad \checkmark$$

Case 2: BB^T .

$$(BB^T)^T = (B^T)^T B^T = BB^T. \quad \checkmark$$

Case 3: $B^T AB$.

$$(B^T AB)^T = B^T A^T (B^T)^T = B^T AB.$$

Step 2 uses $A^T = A$. Therefore $(B^T AB)^T = B^T AB$. ■

Problem 1c: Projection Matrix $C = uu^T$

Proof Strategy

- **Part 1:** Expand Cx by associativity; recognize $u^T x$ as a scalar.
- **Part 2:** Use rank-nullity. $\text{rank}(C) = 1 \Rightarrow$ eigenvalues are in $\{0, 1\}$. Find them directly.
- **Part 3:** Two separate computations: transpose test for symmetry; square C directly for idempotence.
- **Pattern:** Idempotent matrices ($C^2 = C$) always model orthogonal projections in Lay. Eigenvalues are always 0 or 1.

Relevant Theorems & Definitions (Lay)

- **Projection formula (Lay §6.2):** $\text{proj}_u x = \frac{u^T x}{\|u\|^2} u$. For unit vector $\|u\| = 1$: $\text{proj}_u x = (u^T x)u$.
- **Rank-Nullity (Lay §4.6):** $\text{rank}(C) + \text{nullity}(C) = n$.
- **Eigenvalue Def (Lay §5.1):** $Cv = \lambda v$ for nonzero v .
- **Def symmetric (Lay §6.4):** $C^T = C$.

Part 1: $Cx = \text{proj}_u x$

Let $u \in \mathbb{R}^n$ be a unit vector and $C = uu^T$.

$$Cx = (uu^T)x = u(u^T x) = (u^T x)u = \text{proj}_u x,$$

where $u^T x \in \mathbb{R}$ is a scalar (associativity allows reordering), and the last equality is the projection formula with $\|u\|^2 = 1$. ■

Part 2: Eigenvalues of C

$C = uu^T$ has $\text{rank}(C) = 1$ (outer product of two nonzero vectors). By rank-nullity, $\text{nullity}(C) = n - 1$.

$\lambda = 1$, **eigenvector** u :

$$Cu = (uu^T)u = u(u^T u) = u \cdot 1 = u \implies Cu = 1 \cdot u.$$

$\lambda = 0$, **eigenspace** $u^\perp$: For any x with $u^T x = 0$:

$$Cx = u(u^T x) = u \cdot 0 = \mathbf{0}.$$

This null space has dimension $n - 1$.

Eigenvalues: $\lambda = 1$ (multiplicity 1), $\lambda = 0$ (multiplicity $n - 1$). ■

Part 3: C is Symmetric and $C^2 = C$

Symmetry:

$$C^T = (uu^T)^T = (u^T)^T u^T = uu^T = C. \quad \checkmark$$

Idempotence:

$$C^2 = (uu^T)(uu^T) = u \underbrace{(u^T u)}_{=1} u^T = uu^T = C. \quad \checkmark \quad \blacksquare$$

Problem 2: A PD $\Rightarrow A^{-1}$ PD

Proof Strategy

- Use the Spectral Theorem: A symmetric $\Rightarrow A = PDP^T$, P orthogonal.
- Express $A^{-1} = PD^{-1}P^T$; eigenvalues of A^{-1} are $1/\lambda_i$.
- Change variables $y = P^T x$; since P is orthogonal, $y = 0 \Leftrightarrow x = 0$.
- The quadratic form reduces to a weighted sum of squares; positivity of $1/\lambda_i$ closes the proof.
- **Pattern:** For any quadratic form PD argument involving a transformed matrix, diagonalize via the Spectral Theorem and change variables.

Relevant Theorems & Definitions (Lay)

- **Spectral Theorem (Lay §7.1):** A symmetric $\Rightarrow A = PDP^T$, P orthogonal, $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.
- **PD Characterization (Lay §7.2, Thm 5):** A PD $\Leftrightarrow$ all eigenvalues $\lambda_i > 0$.
- **Inverse via SVD/Spectral:** $A^{-1} = PD^{-1}P^T$, eigenvalues $1/\lambda_i$.
- **Orthogonal matrix:** $P^{-1} = P^T$, so $P^T x = 0 \Leftrightarrow x = 0$.

Proof

Since A is symmetric, by the Spectral Theorem:

$$A = PDP^T, \quad A^{-1} = PD^{-1}P^T,$$

where $D^{-1} = \text{diag}(1/\lambda_1, \dots, 1/\lambda_n)$.

Since A is positive definite, all $\lambda_i > 0$, so all $1/\lambda_i > 0$.

Let $y = P^T x$. Since P is invertible, $x \neq 0 \Rightarrow y \neq 0$.

$$x^T A^{-1} x = x^T P D^{-1} P^T x = y^T D^{-1} y = \sum_{i=1}^n \frac{y_i^2}{\lambda_i}.$$

Each term $y_i^2/\lambda_i \geq 0$, and since $y \neq 0$, at least one $y_i \neq 0$, so the sum is strictly positive.

Therefore $x^T A^{-1} x > 0$ for all $x \neq 0$: A^{-1} is positive definite. ■

Problem 3: $|\det(A)| = \text{Product of Singular Values}$

Proof Strategy

- Write A in SVD form: $A = U\Sigma V^T$.
- Take det of both sides; use multiplicativity of det.
- Use the fact that orthogonal matrices have $|\det| = 1$.
- What remains is $|\det(\Sigma)| = \prod \sigma_i$.
- **Pattern:** Whenever a determinant identity involves orthogonal matrices, their determinants cancel to ± 1 under absolute value. SVD is the standard decomposition for relating det to singular values.

Relevant Theorems & Definitions (Lay)

- **SVD (Lay §7.4):** Any $m \times n$ matrix $A = U\Sigma V^T$, U orthogonal ($m \times m$), V orthogonal ($n \times n$), Σ diagonal with $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.
- **Det multiplicativity (Lay §3.2, Thm 6):** $\det(AB) = \det(A)\det(B)$.
- **Orthogonal matrices (Lay §6.2):** Q orthogonal $\Rightarrow \det(Q) = \pm 1$.
- **Det of diagonal matrix:** $\det(\Sigma) = \prod_i \sigma_i$.

Proof

Let $A = U\Sigma V^T$ be the SVD of the square matrix A .

$$\det(A) = \det(U\Sigma V^T) = \det(U) \det(\Sigma) \det(V^T).$$

Taking absolute values:

$$|\det(A)| = |\det(U)| \cdot |\det(\Sigma)| \cdot |\det(V^T)|.$$

Since U and V are orthogonal, $|\det(U)| = |\det(V^T)| = 1$.

$$|\det(A)| = |\det(\Sigma)| = \left| \prod_{i=1}^n \sigma_i \right| = \prod_{i=1}^n \sigma_i,$$

where the last equality holds because $\sigma_i \geq 0$ by definition of singular values. ■

Quick Reference: Proof Patterns

Claim Type	Go-To Tool	Key Step
M is symmetric	Transpose test	$(M)^T = M$
M is positive definite	Spectral + change of vars	$y^T D y > 0$
M is a projection	Show $M^2 = M$, $M^T = M$	Idempotence + symmetry
Det/singular value identity	SVD + $ \det(Q) = 1$	Multiplicativity of det
Inner product shift	Transpose + $A^T = A$	$(Ax)^T y = x^T Ay$